CS520: Analysis of algorithms

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From local probabilities to large independent sets

Where we left off

- $G = (V, E)$ has $n > 0$ vertices, degrees $d(u) = |N(u)|$, and maximum degree Δ; it is finite, simple, and undirected
- X is a random independent set from the hard-core model at fugacity $\lambda > 0$:

$$\mathbb{P}(X = I) = \frac{\lambda^{|I|}}{Z_G(\lambda)}, \qquad Z_G(\lambda) = \sum_{J \in \mathcal{I}(G)} \lambda^{|J|}$$

- Since some outcome is at least as large as the average, $\alpha(G) \geq \mathbb{E}|X|$
- Our goal: bound this average by studying one neighborhood at a time

A useful kind of local inequality

- For each vertex u, consider two quantities:

$$\mathbb{P}(u \in X), \qquad \mathbb{E}|X \cap N(u)| = \sum_{v \in N(u)} \mathbb{P}(v \in X)$$

- These measure occupancy probability at u and the sum of the occupancy probabilities over its neighbors
- We want positive constants β, γ, the same for every vertex, such that

$$\beta \mathbb{P}(u \in X) + \gamma \mathbb{E}|X \cap N(u)| \geq 1 \qquad \text{for every } u$$

Intuition

- Find $\beta, \gamma > 0$ such that for all $u \in V$

$$\beta \mathbb{P}(u \in X) + \gamma \mathbb{E}|X \cap N(u)| \geq 1 \qquad \text{for every } u$$

- Intuition: recall that we want to find a large independent set
- Therefore, $\mathbb{P}(u \in X)$ being large, on average over the vertices, is good
- We can't avoid $\mathbb{P}(u \in X)$ being small, but the inequality says that when it is small, the neighbors of u compensate with $\sum_{v \in N(u)} \mathbb{P}(v \in X)$ being large
- Hopefully, on average over the vertices things work out

Summing the vertex terms

- Recall the indicator identity:

$$|X| = \sum_{u \in V} \mathbf{1}_{\{u \in X\}}$$

- Taking expectations gives

$$\sum_{u \in V} P(u \in X) = \mathbb{E}|X|$$

- This uses linearity of expectation; the vertex events need not be independent (in general they are not!)

Summing the neighborhood terms

- Again by linearity,

$$\mathbb{E}|X \cap N(u)| = \sum_{v \in N(u)} \mathbb{P}(v \in X)$$

- Now count how often a fixed vertex v occurs:

$$\sum_{u \in V} \mathbb{E}|X \cap N(u)| = \sum_{u \in V} \sum_{v \in N(u)} \mathbb{P}(v \in X) = \sum_{v \in V} d(v)\mathbb{P}(v \in X) \leq \Delta \mathbb{E}|X|$$

- Each occupied vertex is counted once for each of its neighbors
- This works for irregular graphs too: we only use $d(v) \leq \Delta$

A local-to-global theorem

Theorem. If $\beta \mathbb{P}(v \in X) + \gamma \mathbb{E}|X \cap N(u)| \geq 1$ for every u, then

$$\alpha(G) \geq \mathbb{E}|X| \geq \frac{n}{\beta + \gamma\Delta}.$$

Proof: sum over all vertices and use the preceding identities:

$$n \leq \beta \sum_u \mathbb{P}(v \in X) + \gamma \sum_u \mathbb{E}|X \cap N(u)| \leq (\beta + \gamma\Delta)\mathbb{E}|X|$$

Now: can we find interesting β, γ?

A first bound for every graph

Recall conditional probabilities

- Write $p = \lambda/(1+\lambda)$ and $Y_u = |X \cap N(u)|$
- If $Y_u > 0$, then u cannot be occupied
- If $Y_u = 0$, adding or removing u preserves independence and we already saw the conditional probability identities
- Consequently,

$$\mathbb{P}(u \in X \mid Y_u = 0) = p, \qquad \mathbb{P}(u \in X) = p\mathbb{P}(Y_u = 0)$$

A probability bounded by an expectation

- Y_u is a nonnegative integer, so

$$\mathbb{E}Y_u = \sum_{k=1}^{d(u)} k\mathbb{P}(Y_u = k) \geq \sum_{k=1}^{d(u)} \mathbb{P}(Y_u = k) = \mathbb{P}(Y_u > 0)$$

- This also follows from Markov's inequality, see [DaKa25]
- Therefore

$$\frac{1}{p}\mathbb{P}(u \in X) + \mathbb{E}|N(u) \cap X| \geq \mathbb{P}(Y_u = 0) + \mathbb{P}(Y_u > 0) = 1$$

- We have the desired inequality with $\beta = \frac{1}{p} = 1 + \frac{1}{\lambda}$ and $\gamma = 1$

What the general bound says

- Substitution in the local-to-global theorem gives

$$\mathbb{E}|X| \geq \frac{n}{\Delta + 1 + 1/\lambda}$$

- Since $\alpha(G) \geq \mathbb{E}|X|$ for every λ, letting $\lambda \to \infty$ gives $\alpha(G) \geq \frac{n}{\Delta+1}$
- This recovers the greedy bound, but it also says something about a whole distribution of independent sets

The general bound is sharp

- Take a clique $K_{\Delta+1}$: the only independent sets are the empty set and the $\Delta + 1$ singletons
- Thus

$$Z_G(\lambda) = 1 + (\Delta + 1)\lambda, \qquad \rho_G(\lambda) = \frac{\lambda}{1 + (\Delta + 1)\lambda} = \frac{1}{\Delta + 1 + 1/\lambda}$$

- Disjoint unions of these cliques also attain equality
- Let's look back at the proof and check that at each inequality we have equality when the graph is a (disjoint union of) cliques $K_{\Delta+1}$

What next?

- OK, we gave an outrageously complicated proof of the greedy bound
- Why?
- The point is that this argument easily incorporates new information about the graph, especially when we have information about neighborhoods
- Neighborhoods in complete graphs are cliques
- The opposite extreme is that we ban all edges in neighborhoods, which is the same as making the graph *triangle free*
- We can find much better constants β, γ in this case

Conditioning on a local view

Why the whole neighborhood is not enough

- For $A \subseteq V$, the **induced subgraph** $G[A]$ keeps exactly the vertices in A and the edges between them
- We would like to analyze $X \cap N(u)$ using $G[N(u)]$
- But an occupied vertex outside $N(u)$ can block vertices inside it
- The right local graph depends on which outside vertices are occupied
- First fix that outside configuration; then compute a smaller hard-core distribution

Revealing the exterior

- Fix any region $A \subseteq V$ and reveal $S = X \setminus A$
- Let $N(S) = \bigcup_{v \in S} N(v)$
- Vertices in $A \cap N(S)$ are forced to be absent
- The remaining graph is

$$F_S = G[A \setminus N(S)]$$

- Conditional on this particular exterior S, possible interiors are exactly the independent sets of F_S

Recall the spatial Markov property, by cancellation



- We will focus on the case when $A = N(u)$ for a vertex u
- The setup: reveal $S = X \setminus N(u)$ and look at what happens in $N(u)$
- Under this setup we observe precisely the hard-core model on the graph F_S formed from $G[N(u)]$ by deleting vertices blocked by S

Neighborhood computations

- The empty set is a valid independent set in F_S with weight 1, so

$$\mathbb{P}(X \cap N(u) = \emptyset \mid S) = \frac{1}{Z_{F_S}(\lambda)}$$

- Recycling the formula for the expected size of an independent set from the hard-core model we have

$$\mathbb{E}[|X \cap N(u)| \mid S] = \frac{\sum_{J \in \mathcal{I}(F_S)} |J| \lambda^{|J|}}{Z_{F_S}(\lambda)} = \frac{\lambda Z'_{F_S}(\lambda)}{Z_{F_S}(\lambda)}$$

- Recall $Z'_F(\lambda)$ means the derivative with respect to λ

Averaging over the exterior

- The hard-core model gives $X \setminus N(u)$ some conditional probability distribution
- So letting S be the *random* independent set we see on the exterior in this case the graph F_S we get is random
- For convenience we will just call this random graph F
- We can now do some gnarly conditional probability calculations:

$$\mathbb{P}(X \cap N(u) = \emptyset) = \sum_S \mathbb{P}(X \setminus A = S) P(X \cap N(u) = \emptyset \mid S)$$

$$= \sum_S \mathbb{P}(X \setminus N(u) = S) \frac{1}{Z_{F_S}(\lambda)} = \mathbb{E}\left[\frac{1}{Z_F(\lambda)}\right]$$

More conditional probability

- For the neighborhood term:

$$\mathbb{E}|X \cap N(u)| = \mathbb{E}\left[\frac{\lambda Z_F'(\lambda)}{Z_F(\lambda)}\right]$$

- The expectations on the right average over possible local graphs

Proving something stronger for convenience

- Then over some random $F \subset G[N(u)]$ the inequality we want becomes

$$\beta p \mathbb{E}\left[\frac{1}{Z_F(\lambda)}\right] + \gamma \mathbb{E}\left[\frac{\lambda Z_F'(\lambda)}{Z_F(\lambda)}\right] \geq 1$$

- The *distribution* of F is tricky, but if for *every* $F \subset G[N(u)]$ we have

$$\beta p \frac{1}{Z_F(\lambda)} + \gamma \frac{\lambda Z_F'(\lambda)}{Z_F(\lambda)} \geq 1$$

then by linearity of expectation we are done

Algorithms need it anyway

- For convenience we prove the mathematically stronger set of inequalities that over all vertices u and induced subgraphs $F \subset G[N(u)]$ we have

$$\beta p \frac{1}{Z_F(\lambda)} + \gamma \frac{\lambda Z_F'(\lambda)}{Z_F(\lambda)} \geq 1$$

- This is no extra work in practice, and it turns out that our **algorithm** will need the stronger set of inequalities anyway

Example: a random neighborhood in a path

- Take the path $a - u - b - c$ at $\lambda = 1$; there are eight independent sets
- For $N(u) = \{a, b\}$, the exterior is $\{u, c\}$ and any subset of it is independent

Exterior S	Compatible interiors	Probability	Vertices of F_S
$\emptyset$	$\emptyset, \{a\}, \{b\}, \{a, b\}$	$4/8$	$\{a, b\}$
$\{c\}$	$\emptyset, \{a\}$	$2/8$	$\{a\}$
$\{u\}$	$\emptyset$	$1/8$	$\emptyset$
$\{u, c\}$	$\emptyset$	$1/8$	$\emptyset$

Example: checking the averages

- Every possible F here is edgeless, with $y = 0, 1, 2$ vertices
- At $\lambda = 1$, $Z_F = 2^y$ and $\lambda Z_F' / Z_F = y/2$
- The formulas give

$$\mathbb{P}(u \in X) = \frac{1}{2}\left(\frac{2}{8} \cdot 1 + \frac{2}{8} \cdot \frac{1}{2} + \frac{4}{8} \cdot \frac{1}{4}\right) = \frac{1}{4}$$

$$m_u = \frac{2}{8} \cdot 0 + \frac{2}{8} \cdot \frac{1}{2} + \frac{4}{8} \cdot 1 = \frac{5}{8}$$

- Direct enumeration agrees: u appears in two sets, a in three, and b in two

Local occupancy

The definition

Fix $\lambda, \beta, \gamma > 0$. The hard-core model on G has **local (β, γ)-occupancy** if, for every vertex u and every induced subgraph F of $G[N(u)]$,

$$\beta \frac{\lambda}{1+\lambda} \frac{1}{Z_F(\lambda)} + \gamma \frac{\lambda Z_F'(\lambda)}{Z_F(\lambda)} \geq 1.$$

- This includes the graph $\emptyset$ with no vertices: $Z_\emptyset = 1$ and $Z_\emptyset' = 0$
- The first term measures the contribution from an empty neighborhood; the second measures expected occupancy inside it
- Checking **every** induced F covers every possible exterior configuration

Why this definition implies our local inequality



- The random F from the conditioning argument is always an induced subgraph of $G[N(u)]$
- Average the defining inequality over its possible values:

$$1 \leq \mathbb{E}\left[\beta \frac{p}{Z_F(\lambda)} + \gamma \frac{\lambda Z_F'(\lambda)}{Z_F(\lambda)}\right]$$
$$= \beta \mathbb{P}(u \in X) + \gamma \mathbb{E}|N(u) \cap X|$$

- The local-to-global theorem now gives

$$\mathbb{E}|X| \geq \frac{n}{\beta + \gamma\Delta}$$

Triangle-free graphs

What does excluding triangles buy us?



- A graph is **triangle-free** if it has no three pairwise adjacent vertices
- If two neighbors of u were adjacent, they would form a triangle with u
- Thus $G[N(u)]$ is edgeless, as is every induced subgraph F of it
- Such an F is determined, for our calculations, by its number k of vertices
- Instead of analyzing many different neighborhood graphs, we only have to check the integers $0 \leq k \leq d(u)$

The local hard-core calculation becomes explicit

- Every subset of an edgeless k-vertex graph is independent, so

$$Z_F(\lambda) = (1+\lambda)^k, \qquad \frac{\lambda Z_F'(\lambda)}{Z_F(\lambda)} = pk$$

- Local occupancy therefore asks for

$$p\left(\beta(1+\lambda)^{-k} + \gamma k\right) \geq 1 \qquad (0 \leq k \leq \Delta)$$

- The first term decreases with k; the second increases
- Conditional on the exterior, each available vertex is selected independently with probability p. Before conditioning, these events can be dependent

A first triangle-free certificate

- First set $\lambda = 1$, so the condition becomes

$$\frac{\beta}{2^{k+1}} + \frac{\gamma k}{2} \geq 1$$

- Choose a real threshold $t > 0$ and set $\beta = 2^{t+1}$, $\gamma = 2/t$
- If $k \leq t$, the first term is at least 1
- If $k \geq t$, the second term is at least 1
- In either case the inequality holds, giving

$$\beta + \Delta\gamma \leq 2^{t+1} + 2\Delta/t$$

A logarithmic improvement

- For $\Delta \geq 2$, take $t = \frac{1}{2}\log_2 \Delta$:

$$\beta + \Delta\gamma \leq 2\sqrt{\Delta} + 4\Delta/\log_2 \Delta$$

- Using $\log_2 \Delta \leq 2\sqrt{\Delta}$ and $\alpha(G) \geq \frac{n}{\beta+\Delta\gamma}$ gives

$$\alpha(G) \geq \frac{n \log_2 \Delta}{8\Delta}$$

- This improves the order of the greedy guarantee by a factor of $\log \Delta$ as Δ grows

Refining the triangle-free bound (optional)

Optimized parameters

- Recall that we want, for $0 \leq k \leq \Delta$ that $p\left(\beta(1+\lambda)^{-k} + \gamma k\right) \geq 1$
- Elementary calculus means we can solve this minimization problem over $k \in \mathbb{R}$
- Write $g(k) = p\left(\beta(1+\lambda)^{-k} + \gamma k\right)$
- Happily, $g''(k) = \frac{\beta p}{(1+\lambda)^k} \log^2(1+\lambda) > 0$ so g is strictly convex
- Then it is minimized at the unique stationary point k^* such that $g'(k^*) = 0$
- We're good at calculus and get

$$k^* = \frac{\log(\beta \log(1+\lambda)/\gamma)}{\log(1+\lambda)}$$

Continuing

- We want $g(k^*) = 1$ but we have two parameters β, γ to choose so we need another equation
- Our final answer is best when $\beta + \gamma\Delta$ is minimized
- So we first solve for β in $g(k^*) = 1$ to get (argh)

$$\beta = \frac{\gamma(1+\lambda)^{1/(p\gamma)}}{e\log(1+\lambda)}$$

- Now we plug this into $\beta + \Delta\gamma$ and minimize over all $\gamma > 0$…

$$h(\gamma) := \beta + \Delta\gamma = \frac{\gamma(1+\lambda)^{1/(p\gamma)}}{e\log(1+\lambda)} + \gamma\Delta$$

More calculus

- More calculus reveals that $h''(\gamma) = \frac{\vartheta e^{\vartheta}}{e\gamma^2 p} > 0$, where $\vartheta = \log(1+\lambda)/(\gamma p)$
- So h is convex and minimized at the unique point γ^* where $h'(\gamma^*) = 0$
- This is doable by hand, but we forgive reaching for a symbolic calculator
- The tricky part is solving equations of the form $z = xe^x$
- You can't really do this except by inventing a name for the inverse function
- We say that $W(z)$ is the positive value such that $z = W(z)e^{W(z)}$
- Then (obviously…)

$$\gamma^* = \frac{\log(1+\lambda)}{p} \frac{1}{1 + W(\Delta \log(1+\lambda))}$$

Putting it all together

- We get

$$\beta + \gamma\Delta = \frac{1+\lambda}{\lambda}\frac{\Delta\log(1+\lambda)}{W(\Delta\log(1+\lambda))}$$

- Is this good?
- As $z \to \infty$ we have $W(z) \sim \log z$
- So if λ is fixed we get as $\Delta \to \infty$ that

$$\beta + \gamma\Delta = \frac{(1+\lambda)\log(1+\lambda)}{\lambda}\frac{\Delta}{\log(\Delta)}$$

Choosing the fugacity too

- Surprisingly, if we also take $\lambda \to 0$ slowly, we can optimize the constant factor
- As $\lambda \to 0$ we have

$$\frac{(1+\lambda)\log(1+\lambda)}{\lambda} \to 1$$

- Our final result is that for n-vertex triangle-free graphs of maximum degree Δ we have (as $\Delta \to \infty$)

$$\alpha(G) \geq (1 - o(1))\frac{n \log \Delta}{\Delta}$$

- There are several proofs of this bound, but none of them surpass the leading constant 1 (see e.g. [Shea83] [DJPR17])

How good is the bound?

- Apart from the lower-order terms, nobody has done better
- We have constructions of graphs such that $\alpha(G) \leq (2 + o(1))\frac{n \log \Delta}{\Delta}$
- The threshold $n \log \Delta / \Delta$ corresponds to a tricky physics phenomenon known as **shattering** and we essentially have no idea how to get past it for this problem

Reading

- Background reading: [DaKa25]
- Theorems 4.1 and 4.2 treat general and triangle-free graphs
- The original paper with essentially this argument is [DJPR17]
- Subsequent work has expanded local occupancy into a framework for fractional coloring and list coloring of graphs too
- Shearer's much older paper uses a greedy approach [Shea83], which is also very interesting!

References

[DaKa25] Davies, E., Kang, R. J. (2025). *The hard-core model in graph theory*. URL: https://arxiv.org/html/2501.03379v2

[DJPR17] Davies, E., Jenssen, M., Perkins, W., Roberts, B. (2017). *On the average size of independent sets in triangle-free graphs*. Proceedings of the American Mathematical Society, 146 (1), 111–124. DOI: https://doi.org/10.1090/proc/13728

[Shea83] Shearer, J. B. (1983). *A note on the independence number of triangle-free graphs*. Discrete Mathematics, 46 (1), 83–87. DOI: https://doi.org/10.1016/0012-365X(83)90273-X